STA2002 Tutorial 5

Part I

1. Let $X_1, X_2, \ldots, X_m \sim N(\mu_1, \sigma_1^2)$ and $Y_1, Y_2, \ldots, Y_n \sim N(\mu_2, \sigma_2^2)$ be two independent random samples from normal populations. When conducting the test $H_0 : \sigma_1^2 = \sigma_2^2$ vs. $H_1 : \sigma_1^2 \neq \sigma_2^2$, we usually use the test statistic $F = \frac{s_1^2}{s_2^2}$, where

$$s_1^2 = \frac{1}{m-1}\sum_{i=1}^{m}(X_i - \bar{X})^2, \quad \text{and} \quad s_2^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y})^2.$$

Please show that under H_0, $F \sim F_{m-1,n-1}$.
Recall that

$$(m-1)\frac{s_1^2}{\sigma_1^2} \sim \chi^2_{m-1}, \quad \text{and} \quad (n-1)\frac{s_2^2}{\sigma_2^2} \sim \chi^2_{n-1}.$$

Since the samples are independent, s_1^2 and s_2^2 are independent. Therefore,

$$\frac{s_1^2/\sigma_1^2}{s_2^2/\sigma_2^2} \sim \frac{\chi^2_{m-1}/(m-1)}{\chi^2_{n-1}/(n-1)} \sim F_{m-1,n-1}$$

Under $H_0 : \sigma_1^2 = \sigma_2^2$, it follows

$$F = \frac{(m-1)\frac{s_1^2}{\sigma_1^2}/(m-1)}{(n-1)\frac{s_2^2}{\sigma_2^2}/(n-1)} = \frac{s_1^2}{s_2^2} \sim F_{m-1,n-1}$$

The decision rule is to reject H_0 if

$$F = \frac{s_1^2}{s_2^2} < F_{m-1,n-1,\alpha/2}, \quad \text{or} \quad F = \frac{s_1^2}{s_2^2} > F_{m-1,n-1,1-\alpha/2}.$$

The critical values $F_{m-1,n-1,\alpha/2}$ and $F_{m-1,n-1,1-\alpha/2}$ can be found by using $qf(alpha/2, m-1, n-1)$ and qf(1alpha/2, m-1, n-1) in R. Alternatively, Statistical Tables can be used. For the latter you may need to use the identity

2. Two lathes are used to produce the same type of ball bearings, and the diameter of the ball bearings follows a normal distribution. Samples of 8 and 9 products are taken from each lathe, and their diameters are measured as follows:

Lathe A:	15.0	14.5	15.2	15.5	14.8	15.1	15.2	14.8	
Lathe B:	15.2	15.0	14.8	15.2	15.0	15.0	14.8	15.1	14.8

Compare if there is a significant difference in the variances of the ball bearing diameters produced by the two lathes (with a significance level of $\alpha = 0.05$).

Let X and Y be the diameter of ball bearings produced by Lathe A and Lathe B, respectively.

$$s_x^2 = 0.0955, s_y^2 = 0.0261.$$

The test statistic is calculated as:

$$F = \frac{s_x^2}{s_y^2} = 3.6590.$$

Taking a significance level of $\alpha = 0.05$, we consult the F-distribution table to find the critical values: $F_{0.025}(7,8) = \frac{1}{F_{0.975}(8,7)} = \frac{1}{4.90} = 0.2041$ and $F_{0.975}(7,8) = 4.53$. The rejection region is defined as:

$$W = F \geq 4.53 \text{ or } F \leq 0.2041.$$

Since the calculated F value is not in the rejection region, we do not reject the null hypothesis. Therefore, it can be concluded that there is no significant difference in the variances of the ball bearing diameters produced by the two lathes.

3. Consider that in a coin guessing game, Stephen made 60 correct guess out of 100 attempts. Do you think he has some trick to help him guess? (Take $\alpha = 0.05$) We let the probablity for Stephen to get a correct guess as p. Then this problem can be regarded as a hypothesis test problem.

$$H_0 : p = \frac{1}{2} \quad \text{vs} \quad H_1 : p > \frac{1}{2}$$

We denote the times of correct guess in 100 attempts as $\boldsymbol{x}$, then if H_0 holds, we have $\boldsymbol{x} \sim b(100, 0.5)$. As the number of samples are large enough, so we take the test statistic as

$$u = \frac{x - 100 \times 0.5}{\sqrt{100 \times 0.5 \times 0.5}}$$

Under hypothesis H_0, the statistic approximately obeys normal Distribution $\mathcal{N}(0, 1)$, so we have the rejection region as: $W = \{\mu > z_{1-\alpha}\} = \{\mu > 1.645\}$, and we have

$$\begin{aligned} p = P\{x \geqslant 60\} &= P\left\{u \geqslant \frac{60 - 100 \times 0.5}{\sqrt{100 \times 0.5 \times 0.5}}\right\} \\ &= P\{u \geqslant 2\} = 0.0228 \end{aligned}$$

Therefore, we should reject the hypothesis H_0, which means Stephen has some trick on this guessing game.

4. **Theorem:** The test statistic for testing the null hypothesis $H_0 : p_1 - p_2 = 0$ is:

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\hat{p}(1 - \hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

where $\hat{p} = \frac{Y_1 + Y_2}{n_1 + n_2}$ the proportion of "successes" in the two samples combined.

(a) Please prove this theorem.

(b) Time magazine reported the result of a telephone poll of 800 adult Americans. The question posed of the Americans who were surveyed was: "Should the federal tax on cigarettes be raised to pay for health care reform?" The results of the survey were:

Non-Smokers	Smokers
$n_1 = 605$	$n_2 = 195$
$y_1 = 351$ said "yes"	$y_2 = 41$ said "yes"
$\hat{p}_1 = \frac{351}{605} = 0.58$	$\hat{p}_2 = \frac{41}{195} = 0.21$

Is there sufficient evidence at the $\alpha = 0.05$, say, to conclude that the two populations - smokers and non-smokers differ significantly with respect to their opinions?

(a) Recall that: $\hat{p}_1 - \hat{p}_2$ is approximately normally distributed with mean: $p_1 - p_2$ and variance: $\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}$ But, if we assume that the null hypothesis is true, then the population proportions equal some common value p, say, that is, $p_1 = p_2 = p$. In that case, the variance becomes:

$$p(1 - p)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)$$

So, under the null hypothesis, we have that:

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - \overbrace{(p_1 - p_2)}^{0}}{\sqrt{p(1 - p)\left(\frac{1}{r_1} + \frac{1}{r_2}\right)}}$$

follows (at least approximately) $N(0,1)$ distribution. Since we don't know the (assumed) common population proportion p, we can estimate p using:

$$\hat{p} = \frac{Y_1 + Y_2}{n_1 + n_2}$$

the proportion of "successes" in the two samples combined. And, hence, our test statistic becomes:

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

(b) The overall sample proportion is:

$$\hat{p} = \frac{41 + 351}{195 + 605} = \frac{392}{800} = 0.49$$

That implies then that the test statistic for testing:

$$H_0 : p_1 = p_2 \text{ versus } H_0 : p_1 \neq p_2$$

is:

$$Z = \frac{(0.58 - 0.21) - 0}{\sqrt{0.49(0.51)\left(\frac{1}{195} + \frac{1}{605}\right)}} = 8.99$$

The p-value is given by: $2 \times P(z > 8.99) = 2 \times (1 - pnorm(8.99)) \approx 0$. Because p-value $< \alpha = 0.05$, we reject the null hypothesis at the significance level of 0.05.

Part II

In this part, we introduce some R examples for hypothesis testing.
Material: *Examples for Hypothesis Testing*. You can download the files from Blackboard (under the tutorial section).